CS 4530: Fundamentals of Software Engineering

Module 13: Software Engineering & Security

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Learning Objectives for this Module

- By the end of this module, you should be able to:
  - Define key terms relating to software/system security
  - Explain 5 common vulnerabilities in web applications and similar software systems, and describe some common mitigations for each of them.
  - Explain why software alone isn’t enough to assure security
  - Explain at least one way in which security can be incorporated into an agile development process

Outline of this lecture

1. Definition of key vocabulary
2. Some common vulnerabilities, and possible mitigations
3. Getting security right is about people as well as software.

Security: Basic Vocabulary (1)

- Security is a set of non-functional requirements (sometimes called “CIA”):
- Confidentiality: is information disclosed to unauthorized individuals?
- Integrity: is code or data tampered with?
- Availability: is the system accessible and usable?

Security: Basic Vocabulary (2)

- Asset: something of value that is the subject of a security requirement
- Threat: potential condition or event that could compromise a security requirement
- Vulnerability: a characteristic or flaw in system design or implementation, or in the security procedures, that, if exploited, could result in a security compromise

Security: Basic Vocabulary (3)

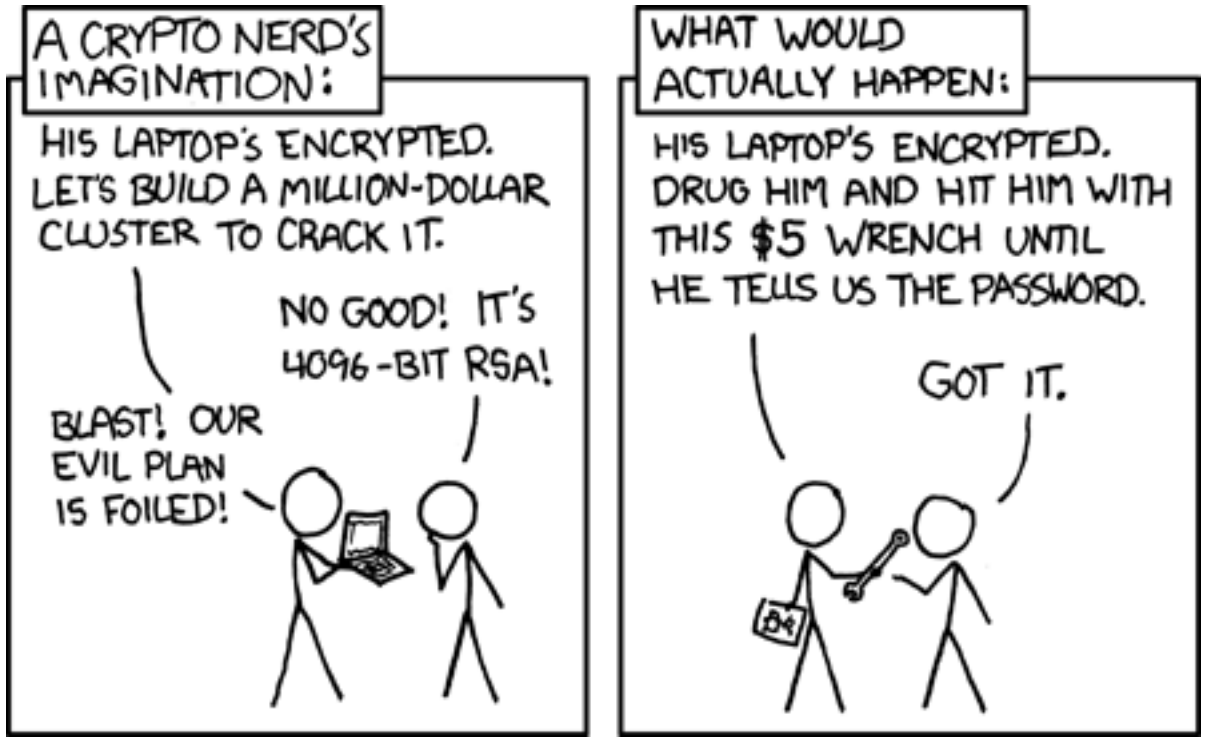
- Attack Surface: a portion of the system that might contain a vulnerability
- Exploit: a technique or method for exploiting a vulnerability
- Attack: an event that results in the realization of a threat
- Mitigation: a technique for making an attack less likely, more expensive, or less valuable to an attacker.

Security is a multiplayer game

The ***threat model*** specifies the rules you imagine that the other person is playing by

- GameNite has radically different applicable threat models than, say, the State Department.

The ***attack surface*** is where the other player gets to play the game



https://xkcd.com/538/

A Simple Threat Model

Simple threat model and attack surface: on the web, an attacker can make arbitrary requests to any API endpoint, send arbitrary messages over websockets, and can directly interpret every message that comes from the server.

Mitigations:

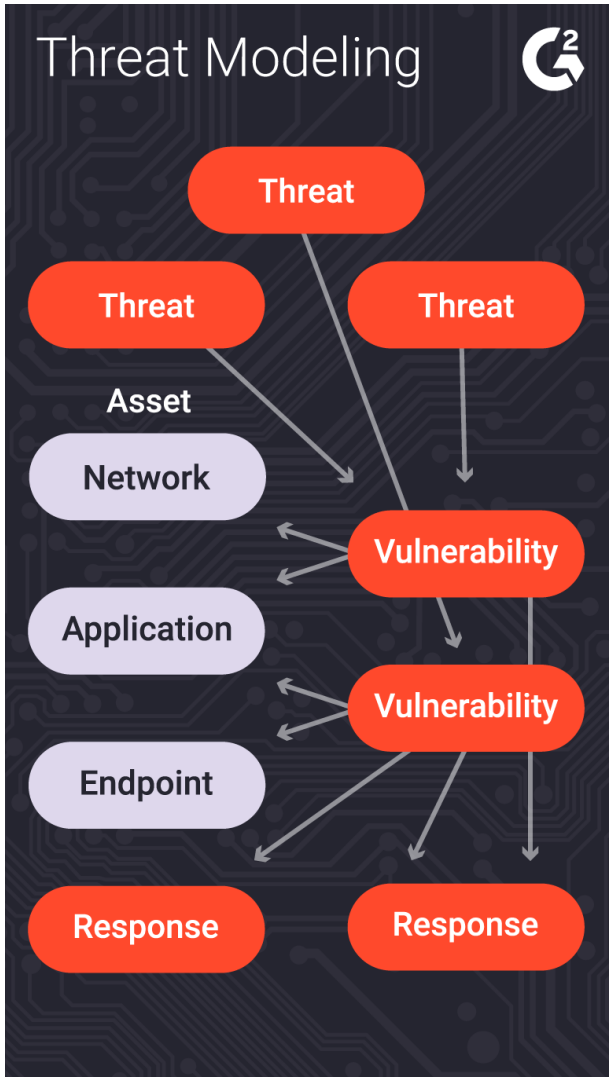
1. Don't put secrets in client code
2. Aggressively validate API requests
   - more on this later.

Security Principle #1: Minimize Attack Surfaces

- Mitigations:
  - Study new APIs and commands for new attack surfaces
  - Remove extraneous dependencies
  - Remove unneeded APIs and commands
  - Authenticate users of APIs

More complex systems require more complex threat models

- What is being defended?
- What malicious actors exist and what attacks might they employ?
- What value can an attacker extract from a vulnerability?
- Who do we trust? What parts of the system do we trust?
- What can we do in case of attack?



STRIDE Framework can help identify Common Threats

- The **STRIDE** framework is a useful categorization of common threats
  - **S**poofing
  - **T**ampering
  - **R**epudiation
  - **I**nformation Disclosure
  - **D**enial of Service
  - **E**levation of Privilege

Security isn't always free

- In software, as in the real world…
- You just moved to a new house, someone just moved out of it. What do you do to protect you belongings/property?
- What are the assets that need protection?
  - residents, furniture, cash, your stuff
- What are the vulnerabilities?
  - doors, windows



Beginnings of a threat model for the new home: what are the costs?

Threat/Exploit	Mitigation
Fire	Smoke alarm, fire extinguisher
Hurricane	Strengthen windows; raise house; move away
Burglary	Burglar alarm? Better locks? Strengthen windows?
Former owner has keys	Change locks
Terrorism	???
Nuclear war	????
???	

Five major classes of vulnerabilities

- Vulnerability 1: Code that runs in an untrusted environment
- Vulnerability 2: Untrusted Inputs
- Vulnerability 3: Bad authentication (of both sender and receiver!)
- Vulnerability 4: Malicious software from the software supply chain
- Vulnerability 5: Failure to apply security policy.

https://owasp.org/Top10/2025/

Vulnerability 1 Example: authentication code in a web application

Curses! Foiled Again!



Front End

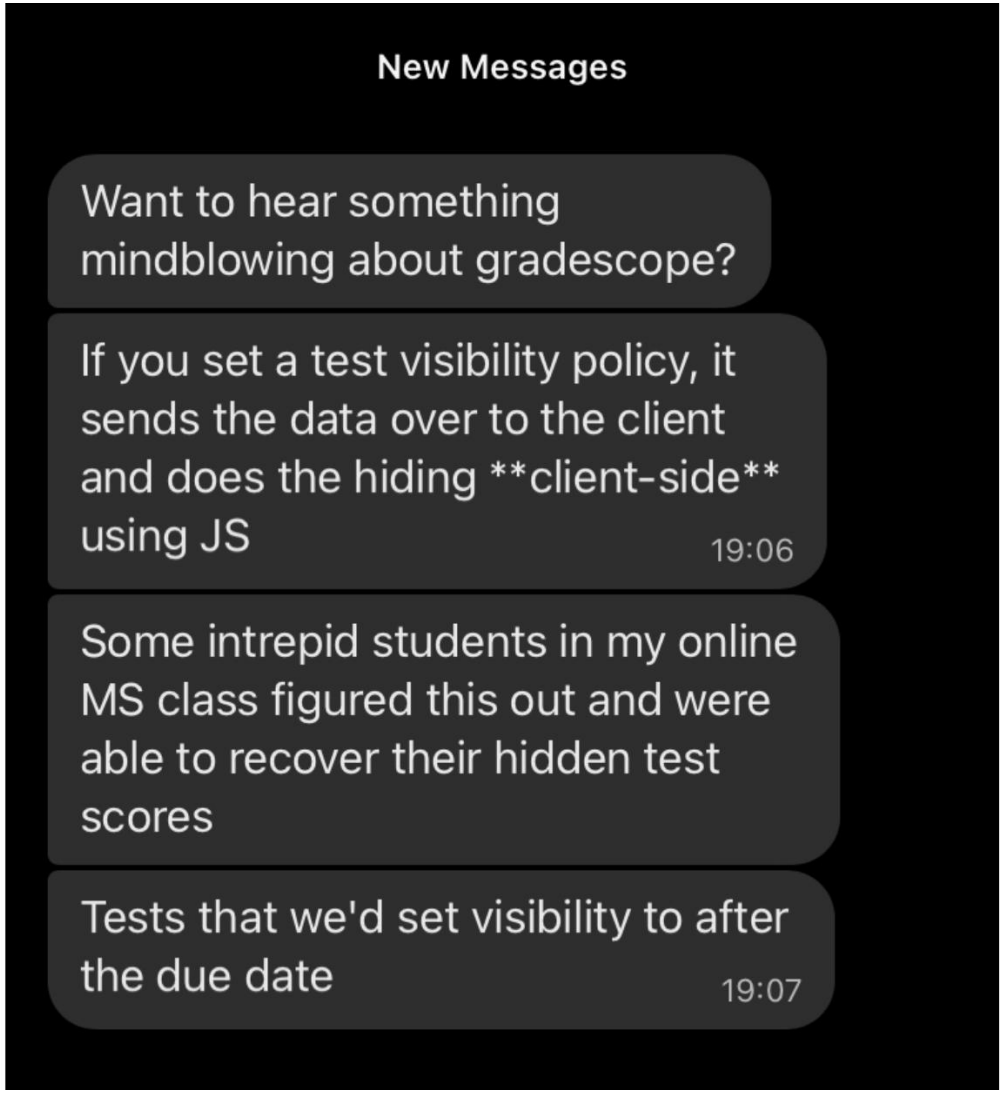
```
function checkPassword(inputPassword:
string){
  if(inputPassword === 'letmein'){
    return true;
  }
  return false;
}
```

Trust boundary

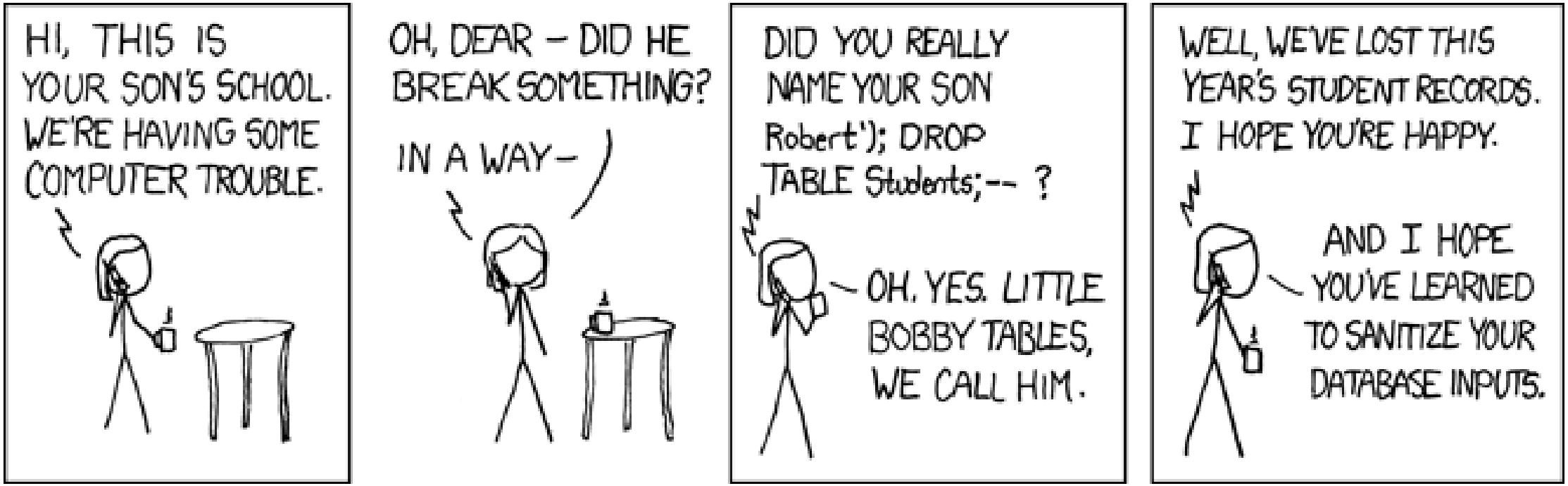
Fix: Move code to back end (duh!)

Back End

Who would do such a silly thing?



Vulnerability 2: Data controlled by a user flowing into our trusted codebase



https://xkcd.com/327/

Example: code injection

```
String query = "SELECT * FROM accounts WHERE
       name='" + request.getParameter("name") + "'";
```

Parameter name	Constructed Query	Effect
Alice	`SELECT * FROM accounts WHERE name='Alice';`	Select a single account
Alice O'Neal	`SELECT * FROM accounts WHERE name='Alice O'Neal';`	SQL Error
5' OR '1'='1	`SELECT * FROM accounts WHERE name='5' OR '1'='1';`	**Select all accounts**

- OWASP https://owasp.org/Top10/2025/A05_20



A code injection attack (in Apache struts) cost Equifax $1.4 Billion



NEWS

Equifax Says Cybersecurity Breach Has Cost $1.4 Billion

EMMA HURT • MAY 10, 2019

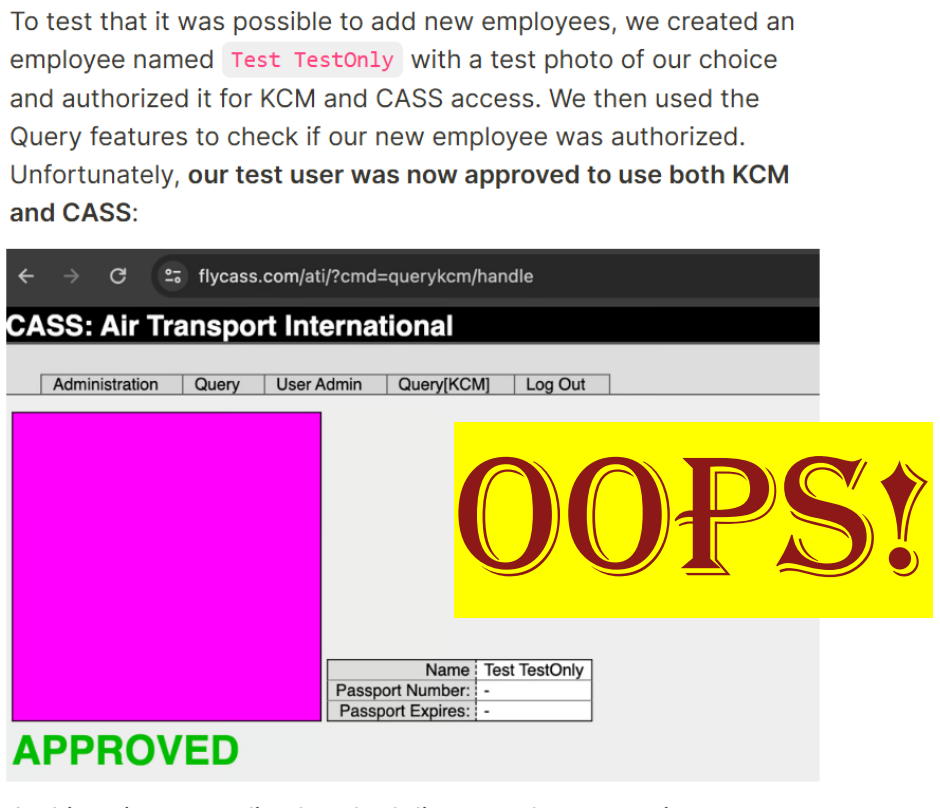
CVE-2017-5638 Detail

Current Description

The Jakarta Multipart parser in Apache Struts 2 2.3.x before 2.3.32 and 2.5.x before 2.5.10.1 has incorrect exception handling and error-message generation during file-upload attempts, which allows remote attackers to execute arbitrary commands via a crafted Content-Type, Content-Disposition, or Content-Length HTTP header, as exploited in the wild in March 2017 with a Content-Type header containing a #cmd= string.

Bypassing airport security via SQL injection (2024!)

- "Known Crewmembers" can get to the cockpit without inspection.
- Large airlines: Each airline runs its own authorization system, but small airlines rely on a vendor
- The authors found one such vendor that had an SQL injection error
- Using the username of ' or '1'='1 and password of ') OR MD5('1')=MD5('1, we were able to login to FlyCASS as an administrator of Air Transport International!



https://ian.sh/tsa

SE-level mitigations for code injection attacks

- Use packages like Zod to sanitize inputs
  - don't do this yourself!
- Use eslint rules to check that Zod is used on every input flow
- Avoid unsafe query languages (e.g. SQL, LDAP, language-specific languages like OGNL in java). Use “safe” subsets instead.
- Avoid use of languages (like C or C++) that allow code to construct arbitrary pointers or write beyond a valid array index
- Never use eval() in JS – executes a string as JS code

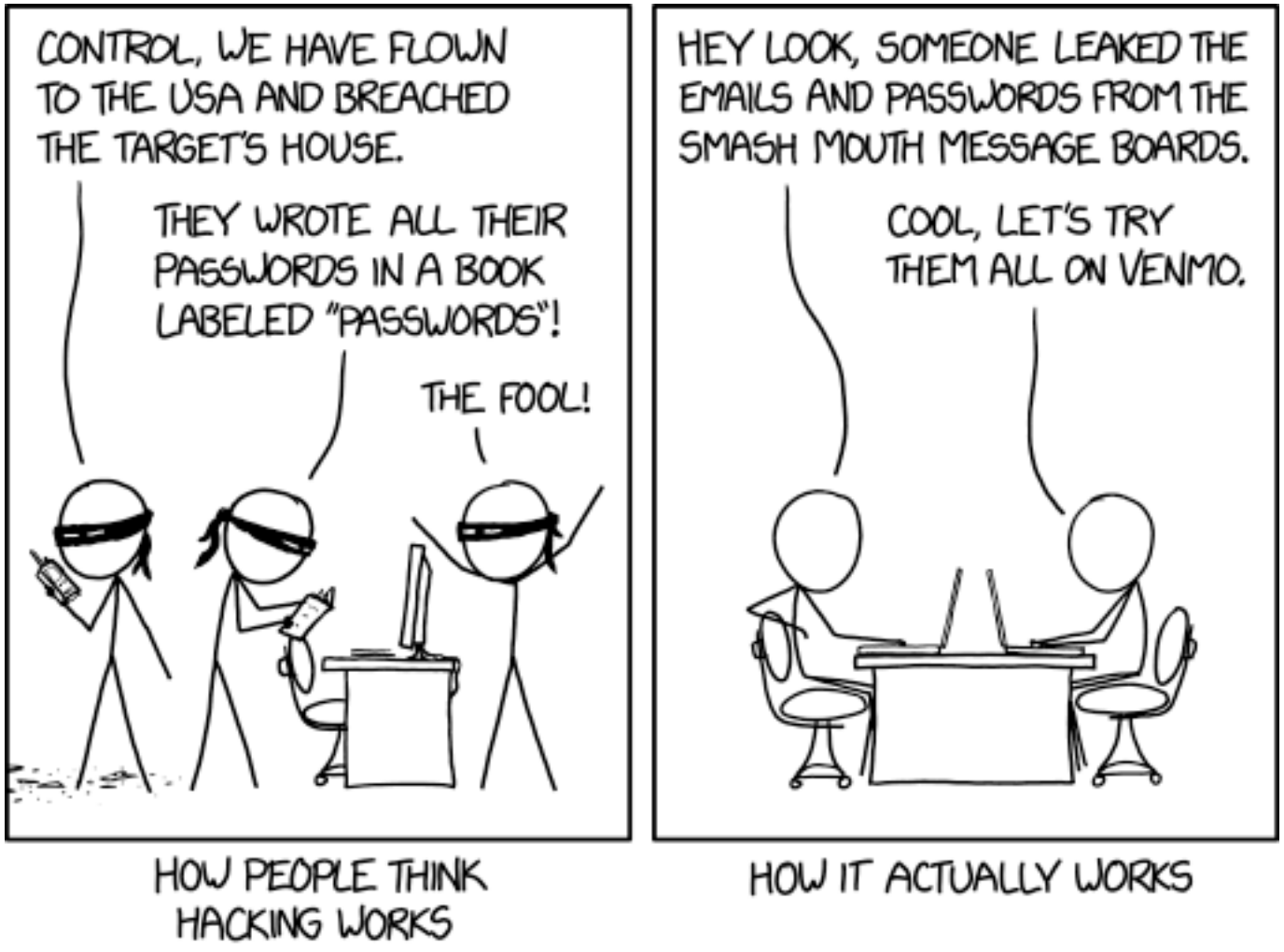
Vulnerability 3: Bad Authentication

- This is mostly about password management
- Attack surface:
  - your password system
  - the attacker may use lists of passwords stolen from other sites

(Inadequate) Mitigation: Don't keep passwords; keep their hashes instead

- Cryptographic hash:
  - Space of hash values is very large (NOT what you'd use for a hash table)
  - hash collisions should be so rare as to be negligible
  - can't go from hash code to original
- Your table consists of (username, savedHash) where savedHash is the hash of the password they created when they signed up for the site.
- Afterwards, the user logs in with (username, maybePassword).
- You compare hash(maybePassword) vs savedHash.

Is this good enough?



…people reuse passwords all the time. Could an attacker take advantage of that?

https://xkcd.com/2176/

Is it safe to store the hashes instead of the passwords themselves?

- Imagine that the attacker has stolen your password db, which looks like
  (username, hashedpassword)
- Next, imagine that the attacker has stolen a table that looks like
  (username, password)
  - this happens all the time!
- So they can easily construct a table that looks like
  (username, password, hash(password))
  - This is sometimes called a "rainbow table"
- And now they can run a join and retrieve a table that looks like
  (username, password)

  for each username that is in both tables



Actual Mitigation: always salt your passwords

- For each user, generate a random number called the "salt"
  - typically 128 bits (16 bytes)
- Your password db should now look like
  - (username, salt, hash(password+salt)
- There's no connection between hash(password+salt) and hash(password)
- So the rainbow table won't work.

Security Principle #2: Don't try this at home; use an established solution instead

- Use established, well-studied crypto packages
- This is a general security principle

```
import { scrypt } from 'node:crypto';
```

```
(alias) function scrypt(password: BinaryLike, salt: BinaryLike, keylen: number, callback: (err: Error | null,
derivedKey: Buffer) => void): void (+1 overload)
import scrypt
```

Provides an asynchronous scrypt implementation. Scrypt is a password-based key derivation function that is designed to be expensive computationally and memory-wise in order to make brute-force attacks unrewarding.

The `salt` should be as unique as possible. It is recommended that a salt is random and at least 16 bytes long. See NIST SP 800-132 for details.

When passing strings for `password` or `salt`, please consider `caveats when using strings as inputs to cryptographic APIs`.

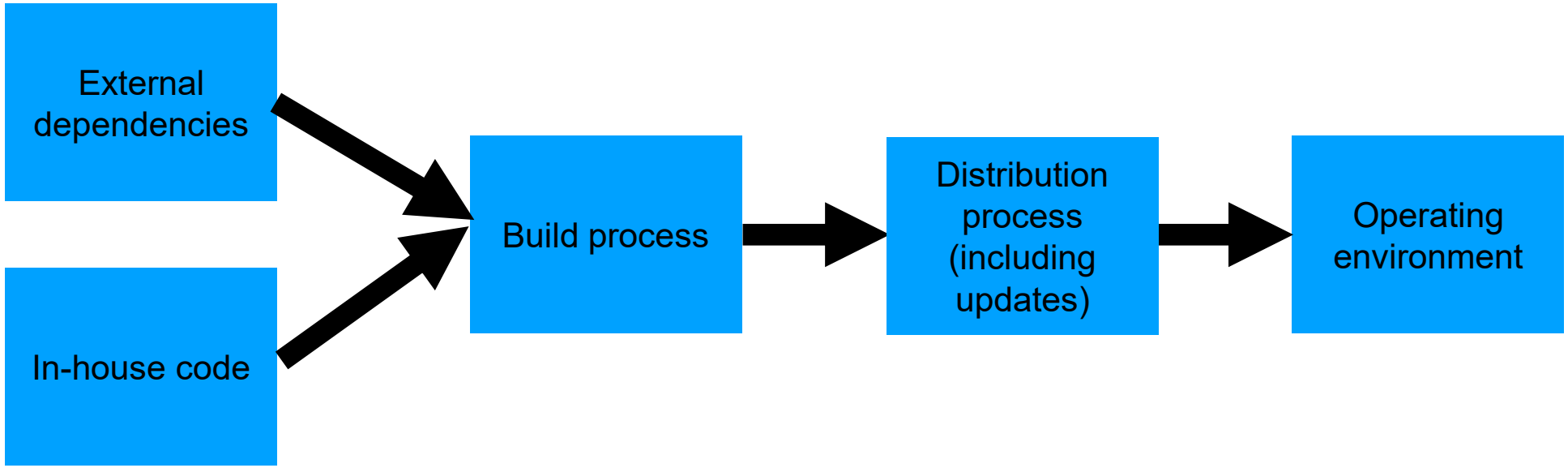
The `callback` function is called with two arguments: `err` and `derivedKey`. `err` is an exception object when key derivation fails, otherwise `err` is `null`. `derivedKey` is passed to the callback as a `Buffer`.

An exception is thrown when any of the input arguments specify invalid values or types.

Vulnerability 4: Supply-Chain Attacks

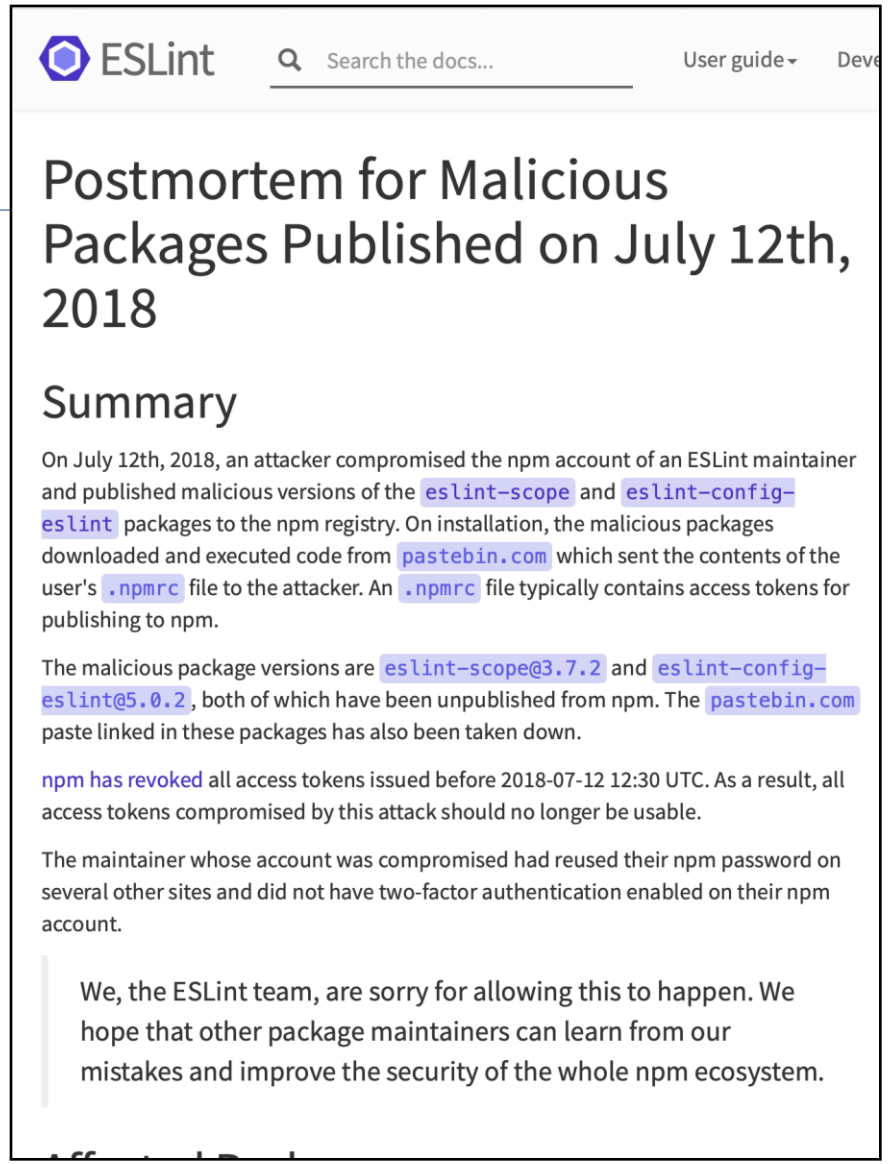
- Our code probably has lots of dependencies.
- Every third-party dependency is a potetial attack surface.

Third-party dependencies show up throughout the development process



Example: the eslint-scope attack (2018)

- On 7/12/2018, a malicious version of eslint-scope was published to npm.
- eslint-scope is a core element of eslint, so many many users were affected.
- Let's analyze this…



ESLint Search the docs... User guide Deve

Postmortem for Malicious Packages Published on July 12th, 2018

Summary

On July 12th, 2018, an attacker compromised the npm account of an ESLint maintainer and published malicious versions of the `eslint-scope` and `eslint-config-eslint` packages to the npm registry. On installation, the malicious packages downloaded and executed code from `pastebin.com` which sent the contents of the user's `.npmrc` file to the attacker. An `.npmrc` file typically contains access tokens for publishing to npm.

The malicious package versions are `eslint-scope@3.7.2` and `eslint-config-eslint@5.0.2`, both of which have been unpublished from npm. The `pastebin.com` paste linked in these packages has also been taken down.

npm has revoked all access tokens issued before 2018-07-12 12:30 UTC. As a result, all access tokens compromised by this attack should no longer be usable.

The maintainer whose account was compromised had reused their npm password on several other sites and did not have two-factor authentication enabled on their npm account.

> We, the ESLint team, are sorry for allowing this to happen. We hope that other package maintainers can learn from our mistakes and improve the security of the whole npm ecosystem.

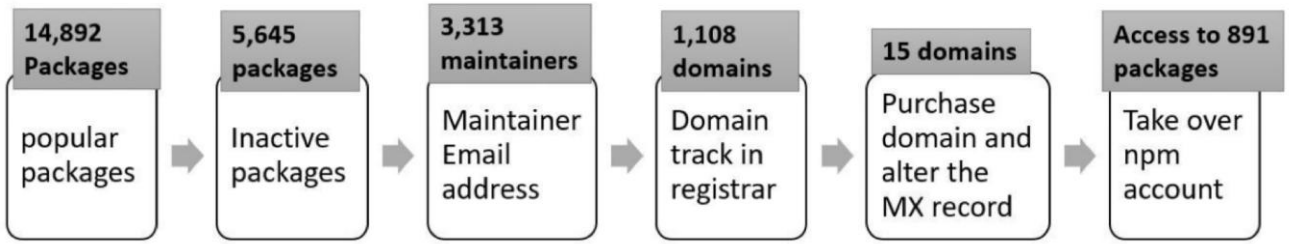
https://eslint.org/blog/2018/07/postmortem-for-malicious-package-publishes/

This incident leveraged several small security failures

- An eslint-scope developer used their same password on another site.
- The other site did not use 2FA
- Password was leaked from the other site.
- Attacker created malicious version of eslint-scope
- Many users did not use package-lock.json, so their packages automatically installed the new (evil) version.
- The malicious version sent copies of the user's .npmrc to the attacker. This file typically contains user tokens.
- Estimated 4500 tokens were leaked and needed to be revoked.

A 2021 NCSU/Microsoft found that many of the top 1% of npm packages had vulnerabilities

- Package inactive or deprecated, yet still in use
- No active maintainers

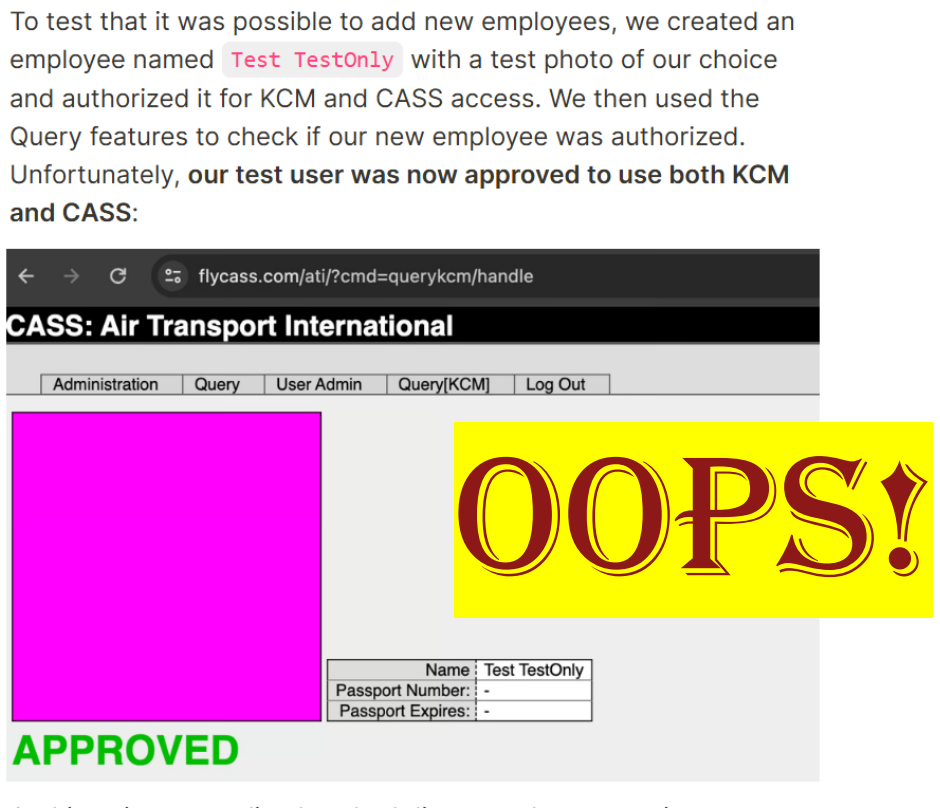


- At least one maintainer with an inactive (purchasable) email domain
- Too many maintainers or contributors to make effective maintenance or code control
- Maintainers are maintaining too many packages
- Many statistics/combinations: see the paper for details.

"What are Weak Links in the npm Supply Chain?" By: Nusrat Zahan, Thomas Zimmermann, Patrice Godefroid, Brendan Murphy, Chandra Maddila, Laurie Williams https://arxiv.org/abs/2112.10165

Your suppliers' risks are your risks

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- Large airlines: Each airline runs its own authorization system, but small airlines rely on a vendor
- The authors found one such vendor that had an SQL injection error
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https://ian.sh/tsa

Your suppliers' risks are your risks.

- MOVEit is a file transfer program owned by Progress Software.
- Over 2500 organizations used the program to move sensitive personal data.
- They were attacked in May 2023.
- Prof. Wand says: my bank didn't use MOVEit, but they used a supplier who did.
- Now, they have to take expensive steps to offer me identity-protection services, etc.

Process-level Threat Mitigations

- External dependencies
  - Audit all dependencies and their updates before applying them
- In-house code
  - Require developers to sign code before committing, require 2FA for signing keys, rotate signing keys regularly
- Build process
  - Audit build software, use trusted compilers and build chains
- Distribution process
  - Sign all packages, protect signing keys
- Operating environment
  - Isolate applications in containers or VMs

Vulnerability #5: Failure to Apply Security Policies

Remember the outline of our lecture:

1. Definition of key vocabulary
2. Some common vulnerabilities, and possible mitigations
3. Getting security right is about people as well as software.

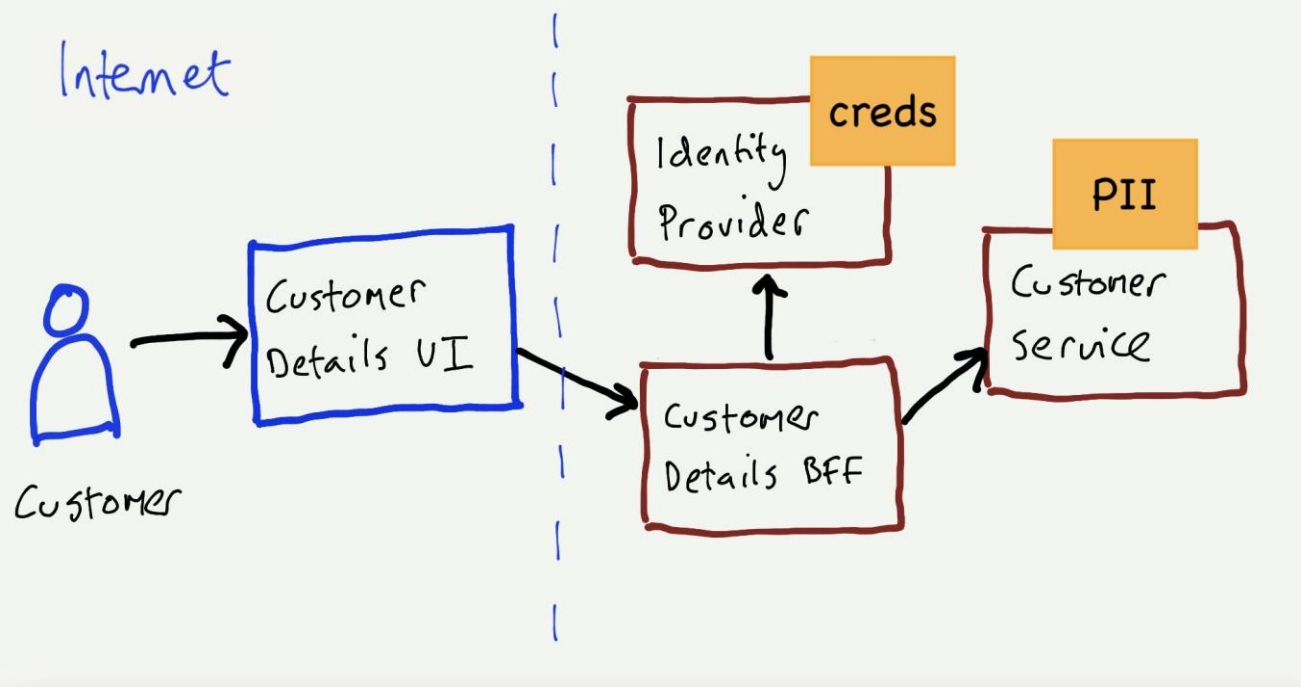
David Blank-Edelman (former head of Systems at Khoury)

"The solution is in front of the screen, not behind it"



Security Principle #3: Every data flow is an attack surface

- There may be vulnerabilities on each flow.
  - Customer -> UI
  - Customer -> Identity Service
  - UI -> BFF
  - BFF -> Customer Service



Systematically review security needs on each attack surface

- Collaborate with stakeholders to understand security needs for each data flow and identify the most vulnerable of them.
- Frequent and small iterations
  - Start with the thinnest slice of the system. e.g.,
    - User registration flow
    - A microservice and its collaborating services
    - Current iteration
  - Repeat and refine them.
- Make this a regular part of the overall QA process

For every data flow, ask “what could go wrong?”

- **S**poofed identity: *Is there Authentication? Should there be?* - Attackers pretending to be legitimate users through stolen credentials, phishing, or social engineering.
- **T**ampering with input: *What about nasty input?* - Attackers modifying data, code, or memory maliciously to break your system's trust boundaries.
- **R**epudiation: *Does the system show who is accountable?* - When something goes wrong, can you prove which user performed an action, or could they plausibly deny responsibility due to insufficient audit trails?
- **I**nformation disclosure: *Is sensitive data inappropriately exposed or unencrypted?* - Unauthorized access to sensitive data through poor access controls, cleartext transmission, or insufficient data protection.
- **D**enial of service: *What if we smash it?* - Attacks aiming at making the system unavailable to legitimate users by flooding or breaking critical components.
- **E**levation of privilege: *Can I bypass Authorization? Move deeper into the system?* - Attackers gaining unauthorized access levels, obtaining higher permissions than intended, or moving laterally through your system.

Choose highest priority vulnerabilities

- Add Highest Priority threats to backlog (as user stories or conditions of satisfaction)

Given the user is logged in
When they request to view their profile page And they have a valid token
Then their profile page is displayed

Given the user is logged in
When they request to view their profile page
But they do not have a valid token
Then they are asked to login or signup

What are you going to do?

- Prioritize and Fix
  - Aim to address manageable number of threats (e.g., 3)
- Examples:
  - Authorization bypass when accessing an API.
  - XSS attack via user input.
  - Denial of service from the internet

- **Repeat and refine**!

A Guide to Threat Modeling by Jim Gumbley: https://martinfowler.com/articles/agile-threat-modelling.html

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